

D501

Machine Learning DevOps

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A data science team at an insurer with 1,100 agents retrains a pricing model by hand each quarter: someone runs notebooks, copies files, and emails operations to deploy. Two of the last four releases broke. The team recently adopted a new office chat tool. What is the core fix?

- A. Send deployment requests through the new chat tool.
- B. Build an automated pipeline that tests, trains, and deploys the model on each change or schedule.
- C. Retrain less often to reduce the number of breaks.
- D. Add a second person to check each manual step.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

An auditor asks a retail bank's ML team to reproduce the exact credit model used last March. The team keeps its code in Git but overwrote the training data file every month. The team's manager is new. What was missing?

- A. More frequent code commits.
- B. A more experienced manager.
- C. Versioning of the training data and model artifacts, tied to each code version.
- D. Version control for the code.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

Which set of items must be versioned together to reproduce a trained model exactly?

- A. Code, training data, configuration and hyperparameters, and environment dependencies.
- B. The saved model file only.
- C. Everything except the environment, which never affects results.
- D. Code only.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A demand forecasting model at a grocery chain with 75 stores shows rising error. The link between price and demand is unchanged, but the chain opened stores in a new region, so the mix of customer demographics in the inputs shifted. The model was built in Python. What is this?

- A. Overfitting.
- B. Concept drift.
- C. Data drift.
- D. A Python version problem.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

Inputs look statistically the same as in training, but after a new law the same customer profile now leads to a different loan default rate. What is this?

- A. Training-serving skew.
- B. Data drift.
- C. No drift, because the inputs are unchanged.
- D. Concept drift.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A fraud model at a payments company with 200 engineers has been live for a year with no monitoring. Confirmed labels arrive 30 days after each transaction. What should the team add?

- A. Retrain every hour no matter what.
- B. Monitor input distributions and prediction rates now, add accuracy tracking as labels arrive, and alert when either shifts.
- C. Monitor only server CPU and uptime.
- D. Start monitoring only after the 30-day labels arrive.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A model works on a data scientist's laptop but fails on the production server because of a different library version. The company recently moved its headquarters to Seattle. What is the standard fix?

- A. Wait until the move to Seattle is complete.
- B. Package the model with its runtime and pinned dependencies in a container image used in every environment.
- C. Install the newest version of every library on the server.
- D. Write the laptop's settings in a README.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

Which problem does a container image NOT solve by itself?

- A. Mismatched library versions between environments.
- B. Differences in operating system packages.
- C. Detecting that model accuracy has dropped because of drift.
- D. Repeatable deployment across servers.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A ride-hailing company with 7 million riders wants to send 5% of live traffic to a new ETA model, compare its error with the current model, and expand only if it performs well. The new model was trained on 18 months of data. Which strategy is this?

- A. Canary release.
- B. Shadow deployment.
- C. Full release to all riders at once.
- D. Blue-green deployment.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

A team wants a new model to receive a copy of every live request and log its predictions, while users are still served only by the old model. Which strategy is this?

- A.** Canary release.
- B.** Blue-green deployment.
- C.** An A/B test.
- D.** Shadow deployment.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A new recommendation model at an online retailer with 90 warehouses cut click-through by 20% after release. The team stores models in a registry with versions and stages. What is the fastest safe response?

- A.** Retrain on this week's data and push straight to production.
- B.** Leave it running a month to confirm the drop.
- C.** Pause shipping at the warehouses.
- D.** Roll back to the previous production version, then investigate.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

An ML pipeline at a utility with 2 million meters trains nightly. Last week an upstream change began sending readings in watts instead of kilowatts, and the pipeline trained and deployed a broken model. The data team has 12 people. What control would have stopped this?

- A.** Automated data validation checks (schema, ranges, units) that fail the pipeline before training.
- B.** Hiring more people for the data team.
- C.** Unit tests on the training code only.
- D.** A larger training cluster.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A data science team at an insurer ~~with 1,100 agents~~ retrains a pricing model by hand each quarter: someone runs notebooks, copies files, and emails operations to deploy. Two of the last four releases broke. ~~The team recently adopted a new office chat tool.~~ What is the core fix?

Target word: core fix **Decides:** Manual, error-prone steps.

A	Camouflage	Changes the channel, not the manual process.
B	Correct	Automation removes manual hand-offs that cause breaks.
C	Inverted operation	Lets the model go stale instead of fixing the process.
D	Adjacent category	Still manual, only slower.

2. Answer C Concept 2

An auditor asks a retail bank's ML team to reproduce the exact credit model used last March. The team keeps its code in Git but overwrote the training data file every month. ~~The team's manager is new.~~ What was missing?

Target word: missing practice **Decides:** Training data was not versioned.

A	Adjacent category	Still no data history.
B	Camouflage	Staffing does not recreate lost data.
C	Correct	Code without its data cannot rebuild the model.
D	Confusion set	They already had it.

3. Answer A Concept 2 · transfer

Which set of items must be versioned together to reproduce a trained model exactly?

Target word: complete set — reverse **Decides:** All inputs to training.

A	Correct	Each one affects the trained result.
B	Confusion set	Keeps the output, not the means to rebuild it.
C	Absolute language	Library versions can change results.
D	Dropped element	Drops data, configuration, and environment.

4. Answer C Concept 3

A demand forecasting model at a grocery chain ~~with 75 stores~~ shows rising error. The link between price and demand is unchanged, but the chain opened stores in a new region, so the mix of customer demographics in the inputs shifted. ~~The model was built in Python.~~ What is this?

Target word: drift type **Decides:** Inputs changed; the relationship did not.

A	Adjacent category	A training issue, not a change over time.
B	Confusion set	The price-demand relationship is unchanged.
C	Correct	The input distribution shifted.
D	Camouflage	The language does not cause shifted inputs.

5. Answer D Concept 3 · transfer

Inputs look statistically the same as in training, but after a new law the same customer profile now leads to a different loan default rate. What is this?

Target word: drift type — reverse case **Decides:** Relationship changed; inputs did not.

A	Adjacent category	A mismatch in feature processing, not a real-world change.
B	Confusion set	The inputs did not shift.
C	Dropped element	Drops the target side of the relationship.
D	Correct	The mapping from inputs to outcome changed.

6. Answer B Concept 4

A fraud model at a payments company ~~with 200 engineers~~ has been live for a year with no monitoring. Confirmed labels arrive 30 days after each transaction. What should the team add?

Target word: monitoring design **Decides:** Labels are delayed 30 days.

A	Absolute language	Blind retraining without signals can make things worse.
B	Correct	Early signals cover the label delay.
C	Adjacent category	System health says nothing about model quality.
D	Dropped element	Drops the early input and prediction signals.

7. Answer B Concept 5

A model works on a data scientist's laptop but fails on the production server because of a different library version. ~~The company recently moved its headquarters to Seattle.~~ What is the standard fix?

Target word: fix for environment mismatch **Decides:** Different dependency versions.

A	Camouflage	The move does not affect library versions.
B	Correct	The same image runs the same everywhere.
C	Adjacent category	Creates a new mismatch with the laptop.
D	Dropped element	Documents but does not enforce the environment.

8. Answer C Concept 5 · transfer

Which problem does a container image NOT solve by itself?

Target word: limit of containers — reverse **Decides:** Containers package environments, not model quality.

A	Inverted operation	Exactly what containers solve.
B	Confusion set	The image bundles these.
C	Correct	That needs monitoring.
D	Adjacent category	A core container benefit.

9. Answer A Concept 6

A ride-hailing company ~~with 7 million riders~~ wants to send 5% of live traffic to a new ETA model, compare its error with the current model, and expand only if it performs well. ~~The new model was trained on 18 months of data.~~ Which strategy is this?

Target word: deployment strategy **Decides:** Small share of real traffic first.

A	Correct	Gradual exposure with a check before expanding.
B	Confusion set	Shadow serves no users.
C	Inverted operation	The opposite of a gradual rollout.
D	Adjacent category	Switches all traffic at once.

10. Answer D Concept 6 · transfer

A team wants a new model to receive a copy of every live request and log its predictions, while users are still served only by the old model. Which strategy is this?

Target word: strategy from the description — reverse **Decides:** No user sees the new model's output.

A	Confusion set	Canary serves real users a share of new results.
B	Adjacent category	Switches users to the new environment.
C	Confusion set	Splits users between models, serving both.
D	Correct	Copies traffic without serving results.

11. Answer D Concept 7

A new recommendation model at an online retailer ~~with 90 warehouses~~ cut click-through by 20% after release. The team stores models in a registry with versions and stages. What is the fastest safe response?

Target word: incident response for a bad model **Decides:** A registry holds the prior production version.

A	Adjacent category	An untested fix during an incident.
B	Dropped element	Drops the need to limit harm.
C	Camouflage	Unrelated to the model.
D	Correct	Restores known-good behavior quickly.

12. Answer A Concept 8

An ML pipeline at a utility ~~with 2 million meters~~ trains nightly. Last week an upstream change began sending readings in watts instead of kilowatts, and the pipeline trained and deployed a broken model. ~~The data team has 12 people.~~ What control would have stopped this?

Target word: preventive control **Decides:** A units change in incoming data.

A	Correct	Out-of-range values would stop the run.
B	Camouflage	Staff size does not create an automated gate.
C	Dropped element	Drops tests on the data.
D	Adjacent category	Speed does not catch bad data.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	6	
Wrong stem number	A figure copied from the prompt that is not the answer.	0	
Adjacent category	The right idea in the wrong bucket.	11	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	3	
Absolute language	Always, never, guarantee — where judgment applies.	2	
Confusion set	Every option is real; the test is exact role-matching.	8	
Camouflage	Answers a stem fact that was never the question.	6	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Automated ML pipelines (CI/CD/CT)	Replace manual training and hand-off steps with an automated pipeline that tests, builds, trains, and deploys on each change or schedule.	Adding more manual reviewers to a manual process.	Notebooks, copied files, and emailed deployments.
2	Versioning for reproducibility	Version code, data, configuration, and environment together so any model can be rebuilt exactly.	Assuming Git on the code alone is enough.	“Reproduce the exact model from last March.”
3	Data drift vs. concept drift	Data drift = the input distribution changes; concept drift = the relationship between inputs and the target changes.	Calling every accuracy drop “concept drift.”	Did the inputs change, or did what they mean change?
4	Production monitoring	Monitor input distributions and prediction patterns right away, and model quality as labels arrive, with alerts that trigger review or retraining.	Monitoring only server uptime, or waiting for labels before watching anything.	A live model with delayed labels and no monitoring.
5	Containers for consistency	A container image packages code with its runtime and pinned dependencies so it runs the same everywhere; it does not monitor model quality.	Expecting containers to fix drift, or fixing version problems by hand.	“Works on my laptop, fails in production.”
6	Deployment strategies	Canary = small share of live traffic first; blue-green = full switch between two environments; shadow = new model sees copies of traffic but serves no users.	Mixing canary and shadow.	“5% of traffic” vs. “users still served only by the old model.”
7	Model registry and rollback	A registry stores versioned models with stages so a bad release can be rolled back fast, then investigated.	Retraining and pushing straight to production during an incident.	A new model hurts a key metric and a registry exists.
8	Data validation tests	Pipelines should test incoming data (schema, ranges, units) and fail before training on bad data.	Testing only the code.	An upstream change in format or units.

Course scope (public description): Machine Learning DevOps focuses on the software engineering fundamentals needed to successfully streamline the deployment of data and machine learning models in a production-level environment. Students will build the DevOps skills required to automate the various aspects and stages of machine learning model building and monitoring over time.